ECE 559 Lab Tutorial 1

Cadence Virtuoso Schematic Editor L Introduction

Contents

1 Introduction

The purpose of the first lab tutorial is to help you become familiar with the schematic editor, *Virtuoso Schematic Composer*. You will create a schematic and a symbol for a static CMOS inverter.

After completion of this tutorial, you should be able to:

- Insert instances into your design
- Connect instances together using wires
- Change instance properties
- Name nets
- Add pins to your design
- Create and edit a symbol cellview
- Check and save your design
- Use buses and tap wires from buses.
- Do a hierarchical design using cells that you have designed

2 Getting Started

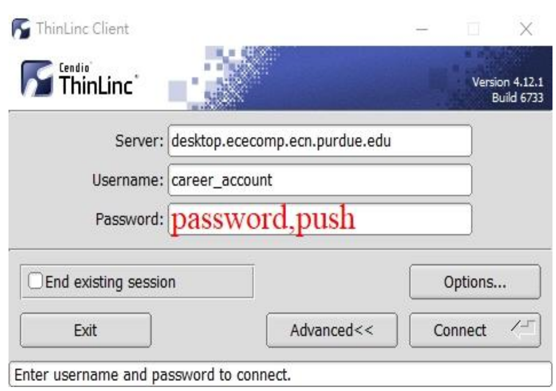
1. Initial Account Connection + Multi-factor Authentication

To get started, download and install ThinLinc Client: https://www.cendio.com/thinlinc/download

Second, open the ThinLinc Client. Enter desktop.ececomp.ecn.purdue.edu in the server field. If you have trouble installing thinlinc client or are using a tablet computer, you can go to https://desktop.ececomp.ecn.purdue.edu in your web browser; however, it is **recommended** that you use the desktop client when possible.

Since Purdue now requires multi-factor authentication, there are some changes that needed to do in order to log in your course account.

Connect to server: **desktop.ececomp.ecn.purdue.edu**



Enter your Purdue login, so username + “password,push” as password. “password” is your full password for your career account, “,push” should be typed exactly like that.

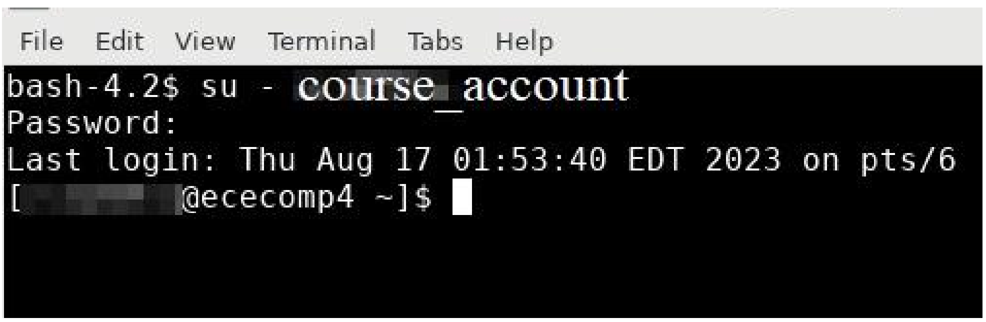
Just like logging in to Brightspace, this requires you to click the notification on your phone. You will get up to **2** login notifications, so just open your phone until you are logged in. For more info (including **VPN** info) on thinlinc: https://engineering.purdue.edu/ECN/Support/KB/Docs/ECEThinlinc

Once you have connected, you will load into a desktop environment. The first time you log in, make sure to select “**Use default config**” in the window that pops up

If you accidentally click “One empty panel,” right click on the desktop and click “open terminal here,” then run `rm -r ~/.config/xfce4` Next, disconnect from thinlinc (F8->Disconnect). Finally, reconnect using thinlinc and check the box for “End existing session”.

Once you logged in, you are in your ECN account NOT your course account, so EDA tools are not able to be used here just yet.

Open a **terminal** and run **“su - course_account”**, where course_account is your course account, then enter the password to finish the login process. Now only this terminal is in your course account. You can run `whoami` to see what account is currently logged in.



There is an advanced way to log in faster, for more info (will need Purdue VPN access to view): https://wiki.itap.purdue.edu/display/ECEEDA/Course+Account+Login+Guide

*Tips. some people that are not familiar with full terminal interface may wonder how to access saved images or files. You can type **nautilus&** in the terminal to pop out a GUI file browser that you are more used to. You can also type **firefox&** to open a web browser where you can upload files directly from your course account. This may come handy when you are trying to export images for writing your report.

2. Account Setup

- Use the opened terminal window.
- Run the command **chsh**, enter your password (you will not see the characters as you type the password), then type **/bin/bash**
- Run the command **passwd** and change your password to a new memorable password.
- Enter **`/package/eda/setup/course_cfgs/ece559.bash`** in your terminal window and press enter. Then press y. Note that you only need to do this once.
- The above command copies the files `.bashrc` and `cds.lib`. The cds.lib file contains the path of the library files and the *.bashrc* file contains the commands used to source the toolset.
- Completely log out and log back in (type **exit** in the terminal or close the terminal window to log out, use the “su – course_account” command to log back in.)

We are ready to start Virtuoso. To run virtuoso, simply type `virtuoso&` (the & symbol tells your shell to run virtuoso in the background).

3 Software Documentation

Note that this tutorial and the following series cover only the very fundamental concepts of creating CMOS schematics, symbols and layouts, simulating circuits, performing layout verification and parasitic extraction from layout using Cadence. Please refer to the software documentation should you require additional information.

- To access the software documentation, go to **Help - > Virtuoso Documentation Library** in the CIW window.

- In the software documentation, more detailed information can be found that you may find helpful.

4 Virtuoso Schematic Editor L Basics

The Virtuoso Schematic Editor L is used to create the schematic of your design. In the schematic, it will contain devices (transistors) connected together with nets (wire connections).

4.1 Launch Virtuoso Schematic Editor L

- Enter `virtuoso &` in the terminal window command prompt. A window should appear with the title *virtuoso* at the top. This window is known as the *Command Interpreter Window* (CIW). It is the main control window for the schematic composer software. Various properties can also be changed in this window.

- The CIW displays a running history of the commands that you execute and their results. It also shows status and error messages from the schematic composer software.

- When commands are run, the CIW will display prompts for action.

4.2 Library Manager

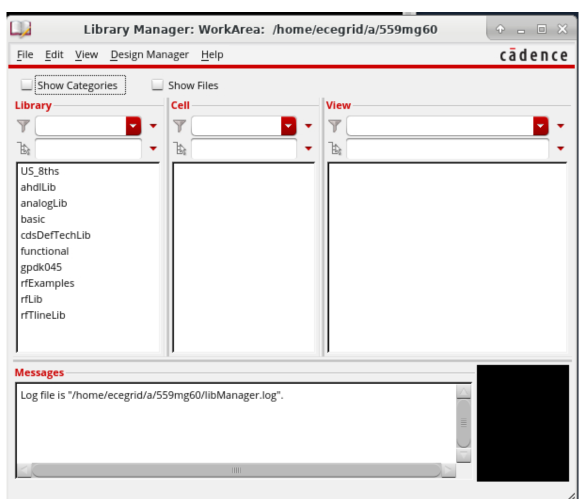
The Library Manager allows you to manage (create, copy, move, delete) libraries. It is recommended that all changes in the libraries be done here to preserve the integrity of all the files associated with your design.

- To run Library Manager, click on **Tools -› Library Manager**.

- Cadence uses the term *library* to mean both reference libraries, which contain defined components for a specific technology, and design libraries, in which you create your own designs. The designs are called *cells*.

- Each cell can have multiple representations, such as a *symbol* or a *schematic*. These representations are called *cellviews*.

- There should be several libraries already present in the *Library* column. Click on any of the libraries. It will list some cells in the *Cell* column that belongs to this library. You should see a library called gpdk045. This will probably be the most often used library for this course, as you will find out later on.



- The cellviews associated with the cell will appear in the *View* column if you click on any of the cells listed.

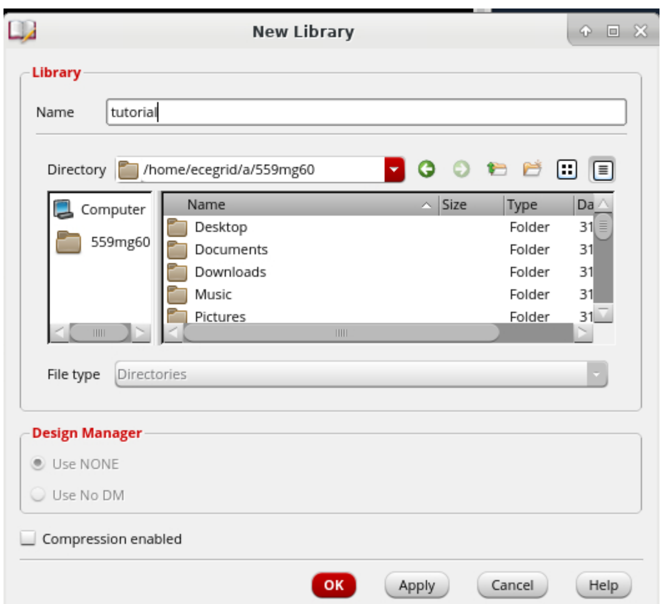
- The pathnames of these libraries are defined in the *cds.lib* file. The paths can be altered manually with a text editor if required.

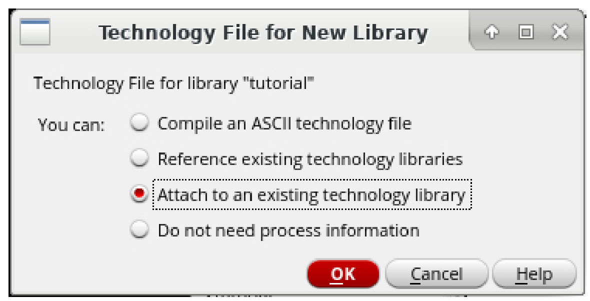
- It is also possible to open designs from the library manager. To do this, either right click on the cellview and select *Open* from the menu that appears or simply double click on the cellview name.

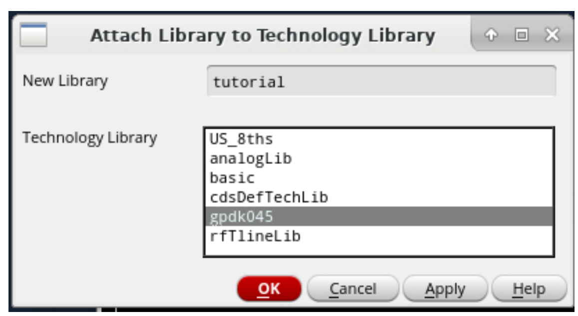
4.3 Creating a New Library

- To create a new library, click on **File -> New -> Library** (you can do this in either the *CIW* window or the Library Manager). A new window called *Create Library* appears and type `tutorial` in the *Name*. If you see a library named `tutorial` already there, you don't need to create it.

- A window called *Technology Library* will open. Click on *Attach to existing tech library*. The technology file (techfile) stores all the instances, process and rules files required for schematic and layout design. Select **gpdk045**. Click **OK**.



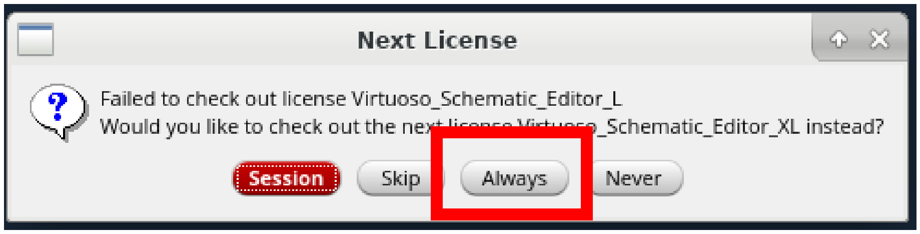




4.4 Creating a New Cellview

- To create a new cell view, first select your new library under *Library* in *Library Manager*, click on **File -> New -> Cell View** (you can do this in either the *CIW* window or the *Library Manager*). A window appears and select *tutorial* for the *Library* field, and type in `inverter` for the *Cell* field. The *View* field should be *schematic* and the *Open with* field should be *Schematics L*. Click **OK** when done.

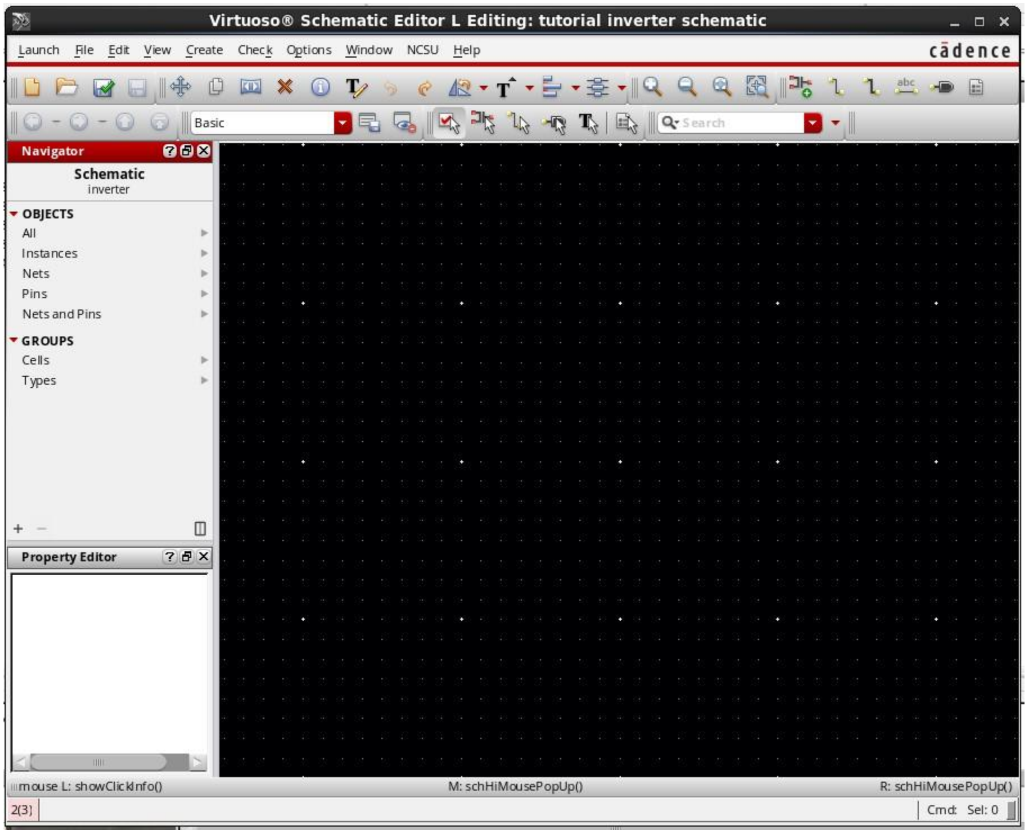
- You may get a License warning. Click “Always” on the below popup:



- A new window named *Virtuoso Schematic Editor L Editing: tutorial inverter schematic* should appear. This is the schematic window or cellview. Note that the last parts of the window name correspond to the library (tutorial) and the cellview (inverter) that you are currently working on.

- Take a look at the command menu and icons on top. Clicking on the top (drop down) menu will reveal more command options. If there is an arrow to the right of the command option in the drop down menu, it means that there are more options under it.

- Next to certain command options, there are some letters next to it. These are bindkeys (or more commonly known as hot-keys) that invoke the command using simple key presses. They will become handy when you get more familiar with the schematic editor and the bindkeys.



- The icons on the top correspond to several most frequently used commands such as add instance, change instance properties, add wire, zoom in, zoom out, undo, delete etc. By placing the mouse cursor on top of the icon, the name of the icon appears.

- A pop up menu appears when you place the cursor on any empty portion of the schematic and press the right or middle mouse button.

5 Drawing a Schematic for a CMOS Inverter

Now you are ready to draw the schematic of a CMOS inverter as shown below in the illustration. From the figure, you can see that the inverter consists of two transistors (one n-type and one p-type), these are known as instances. There are also In, Out, Vdd and Gnd pins.

5.1 Adding Instances

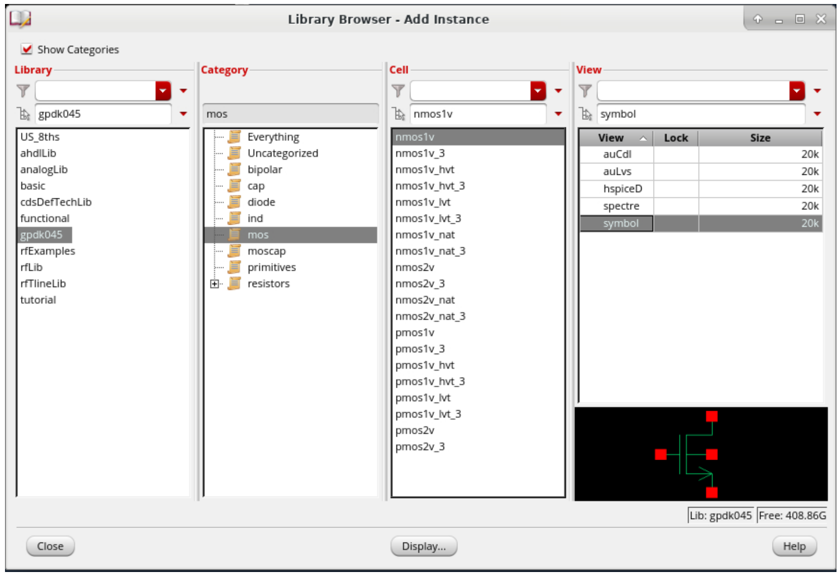
- To add a transistor to your schematic, click on the **Create Instance** icon. Two new windows, one named *Component Browser* and the other named *Add instance* appear. There are several other ways to bring up these windows. One way is to select **Create -> Instance** from the top menu. Another way is to press the right or middle mouse button within the cellview, then select **Add Instance** when the pop-up menu appears. The last but probably most convenient way is to use hotkey (in this case, press 'i' on the keyboard).

- Click *Browse* on the add instance window. In the *Library Browser* window, check the box for "Show Categories", select the *gpdk045* library, then select *mos*. Next, select *nmos1v* then *symbol*. Now move the mouse cursor back to the schematic window and you will find a symbol representing an n-type transistor attached to the cursor.

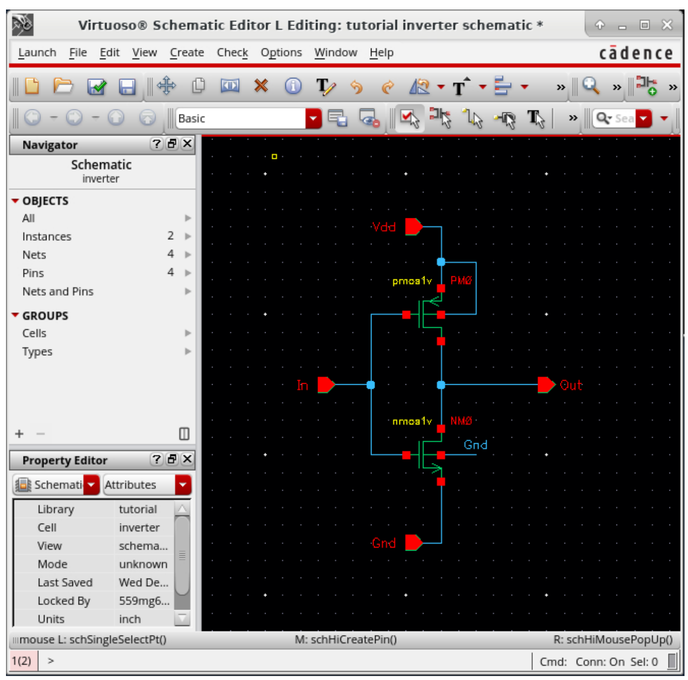
- You can rotate or flip the instance (sideways or upside down) by clicking the **Rotate**, **Sideways** and **Upside Down** buttons in the window before placement.

- If you have accidentally chosen the wrong instance, press the `Esc` key or click the **Cancel** button on the *Add Instance* window.

- Move the cursor to a desired location on the schematic window and click the mouse button to put the transistor in place.



- After the placement of the n-type transistor, it will continue to prompt you to add another instance (same instance by default). This will allow you to place multiple instances onto your cellview. Return to the *Component Browser* and under the same library, select *mos* and then *pmos1v*. Place a p-type transistor in the schematic window.



- Press the `Esc` key to stop adding more instances.

- You can simply click and drag the instance around to re-position them on your cellview.

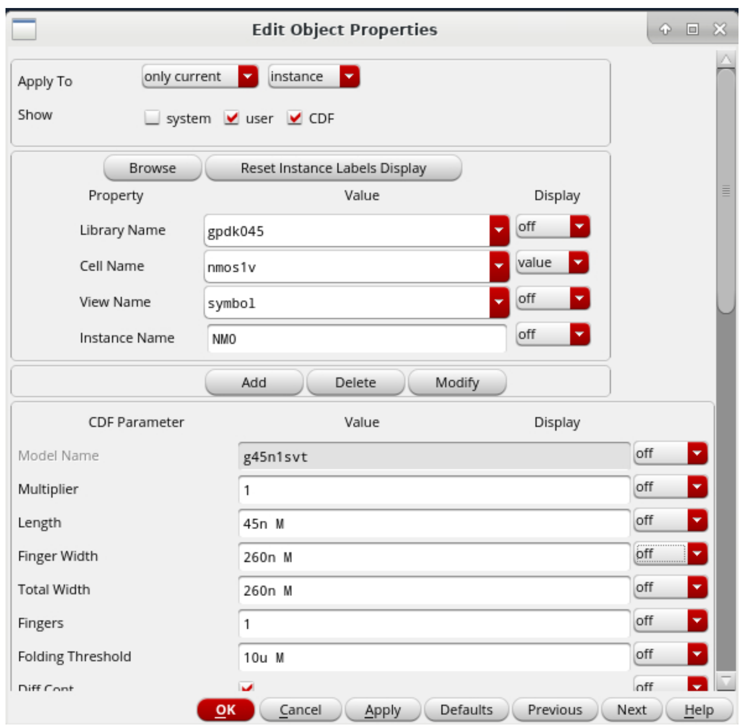
5.2 *Editing Instance Properties*

- To modify the properties of an instance, such as the width and length of the n-type transistor, select the n-type transistor and click the **Edit Properties** icon on the top or selecting **Edit -> Properties -> Objects** from the drop down menu (hot-key is 'q').

- A new window named *Edit Object Properties* appears. The *Library Name*, *Cell Name* and *View Name* are displayed near the middle section. Ensure that these values correspond to the right instance before modifying the properties.

- For the *Instance Name* field, it can be changed to any value for easy identification between instances.

- For the *CDF Parameters* change the *Total Width* to `0.26u` and the *Length* to `45n` (`u` represents micrometers and `n` represents nanometers). Use the `Tab` key or mouse to move between fields and do not press the `Enter` key unless you are done.



- To change the properties for another instance, click on the instance and the *Edit Object Properties* window will be updated to the new instance.

- As an exercise, change the properties of the p-type transistor such that the width is `0.39u` and the length is `45n`.

- After done editing, click **OK** or press `Enter` to quit.

- Note that other instance properties can be edited in the same manner.

5.3 Deleting Instances

- To delete an instance, click on the instance to be deleted. A box around the instance indicates that the instance is selected. Click the delete icon on the top of the schematic window or press the `Delete` key to delete.

- To delete multiple instances, you can first select multiple instances by pressing the `shift` key while selecting the instances with the mouse or by holding the left mouse button down and dragging a box, then execute the delete command as above.

- Another way is to execute the delete command, then select the instances that are to be deleted. Note that the delete command will remain active until you cancel the command by pressing the `Esc` key. This is also true for most other commands.

5.4 Adding Pins

- To add pins (used to connect your current design to external circuits), either click on the **Pin** icon, or select **Create -› Pin** from the drop down menu (hot-key is 'p').

- A new window named *Create Pin* appears.
  - Type `In` for the *Names* field. o Select *input* for the *Direction* tab.
  - Leave other settings at their default state.

- Move the mouse cursor back to the schematic window and you will find a symbol representing a pin attached to the cursor, similar to the placing of an instance.

- You can rotate or flip the symbol (sideways or upside down) by clicking the **Rotate Left**, **Rotate Right**, **Flip Vertical** and **Flip Horizontal** buttons in the window before placement.

- As an exercise, add an output pin named *Out*. The *Direction* tab should be set to `output` this time. Also add the *Vdd* and *Gnd* pins with the *Direction* tab set to `inputOutput`.

5.5 Adding Wires

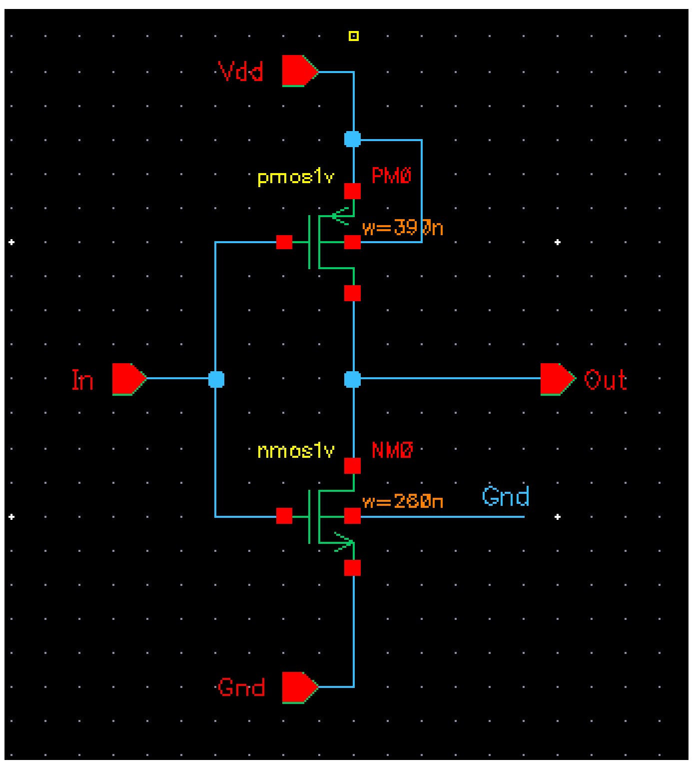
- To add wires to connect the instances together, click on the *Create Narrow Wire* icon on the top. Alternatively, click the right or middle mouse button within the cellview and select *Wire (narrow)* in the pop up. You can also use hot-key ‘w’. You might have noticed that there is a similar icon named *Wire (wide)*. It is used to create buses.

- A new window named *Add Wire* appears. Leave the *Draw Mode* as *route* unless you absolutely need to draw non-rectilinear wires.

- Click (do not hold the left mouse button) on the wire starting point (for example, at the red boxes indicating an instance pin). Move the mouse cursor and click again for each wire segment.

- You might notice that as you move the mouse cursor, a small diamond shape appears over the connection object closest to the pointer. To end the wire, press `s` to snap to the nearest object that shows a diamond shape. Another way is to click a schematic pin, an instance pin, or another wire or double-click on a new wire endpoint.

- Finish up the schematic by connecting the instances together with wires.



You should now have a schematic similar to the one in the diagram (schematic of CMOS inverter).

5.6 Naming Nets

- You might have noticed that in the schematic, the nmos bulk pin is not connected to the schematic. By naming the net (or wire), it is possible to indicate that there is an electrical connection between other nets or pins. This could help in reducing the amount of wire cluttering that makes the schematic hard to read.

- To name a net, select **Create -› Wire Name** from the drop down menu or select **Add Name** from the pop up menu on the wire. You can also use hot-key 'l'. A new window named *Create Wire Name* appears. Type in `Gnd` for the *Names* field and move the mouse cursor to the net that is to be named *Gnd*.

- It is possible to add several names to different nets at one time. To do this, simply type the names separated by a space in the *Names* field and the names will be added in order when you click on the various nets.

6 Checking and Saving Your Schematic

Once you think that your schematic is complete, you will need to run a check on it. This check only checks for very rudimentary problems (such as unconnected pins or dangling wires) and for many more subtle or obscure problems that may cause trouble later on in programs that try to use your schematic.

- To check your schematic, click on the **Check and Save** icon. The results of the check are displayed in the CIW. If everything goes well, you should see a message in the CIW:

```
Schematic check completed with no errors.
“tutorial schematic inverter“ saved.
```

- As an exercise, delete a wire segment from your schematic and do a check. A window named *Schematic Check* will appear indicating the number of errors and warnings found. Click **Close**.

- In the cellview, flashing markers highlight the areas that are causing the errors or warnings.

- To understand the cause of errors, select the drop down menu item **Check -› Find Marker**. A new window appears showing the list of errors and warnings.

Clicking on a particular error and warning from the list will highlight the markers that correspond to that particular error or warning highlighted on the schematic. Click **Delete** to remove the marker from the cellview.

- There might be situations, for example, where you do not need to use a certain pin on an instance and would just want to leave it "floating". The checker will indicate a warning that the pin is floating. To ignore the warning and to prevent it from appearing repeatedly, select the warning from the list and click **Ignore**. That particular warning will not appear in the next check. To make the warning reappear, click **Restore All** in the same window.

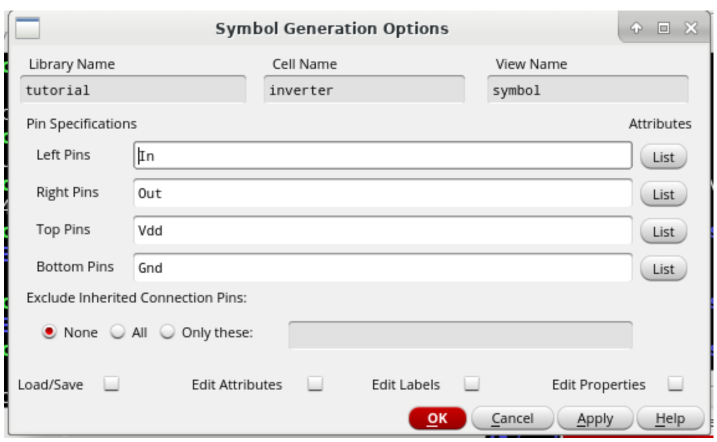
- Fix the schematic, do a check and save it.

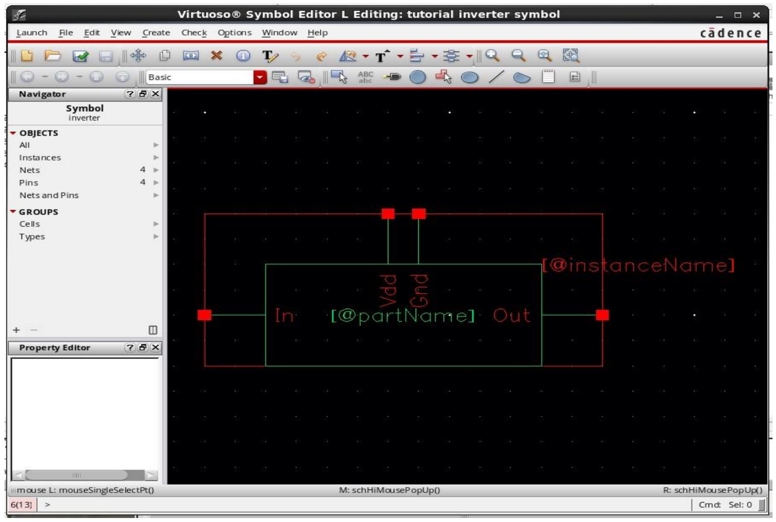
7 Creating and Editing a Symbol for Your Schematic

At this point, you should have a schematic "view" of a CMOS inverter "cell" that passes the check without errors. You will now generate a symbol "view" for the inverter "cell".

- From your cellview of your inverter schematic, select **Create -> Cellview -> From Cellview** from the drop down menu. A new window named *Cellview From Cellview* appears. The window shows entries for the active design, so *Library Name* should be set to *tutorial*, *Cell name* to *inverter*, and *From View Name* to *schematic*. Check to make sure that *To View Name* is set to *symbol* and all the above information is correct, then click **OK**.

- The *Symbol Generation Options* window appears. Edit the lists so that *Vdd* is a top pin, *Gnd* is a bottom pin, *In* is a left pin, and *Out* is a right pin. Click **OK**.





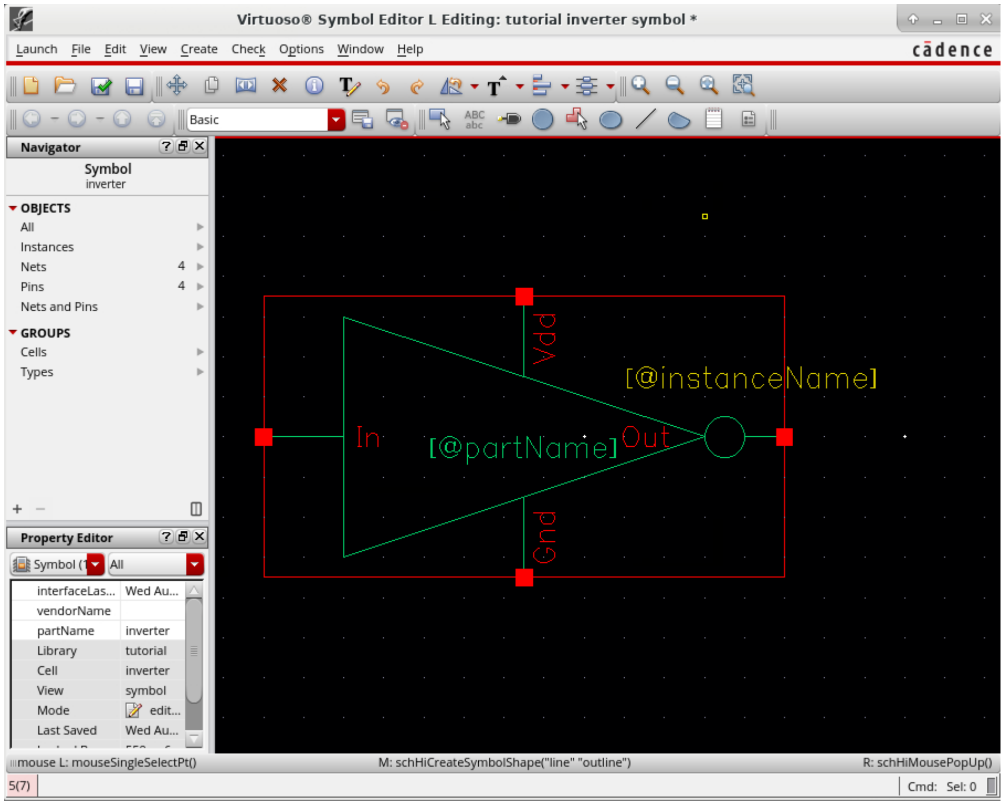
A new window called *Virtuoso Symbol Editor L Editing: tutorial inverter symbol* appears, displaying the *inverter* symbol generated by the schematic editor software. This symbol is functional and can be readily used, but the shape is a simple rectangle and not the conventional inverter symbol.

- Note that the icons on top are slightly different from the ones in the Schematic Editor window.

- To draw the new shape, select **Create -› Shape -› Polygon**. Point to the first point of the polygon, followed by the next point until you end with your original starting point and form a triangle. Add a small circle at the apex of the triangle at the output end by selecting **Create -› Shape -› Circle**. Point and click to define the center of the circle and move to size the circle by moving the mouse.

- Carefully delete the inner box. Do not delete the outside box that runs through the pin squares.

- Select the top and bottom sides of the outside box and move them out to the points on the triangle (click and drag), so that the entire triangle is inside the box. This box determines the boundaries of the symbol (i.e., determines what area you click on the schematic to select the symbol) · The final result should look like this:



Check and save the symbol by using the **File -> Check and Save** command from the drop down menu.

- If you take a look at the *inverter* cell in the *tutorial* library, you can see that it has now a schematic cellview and a symbol cellview.

- Close the Symbol Editor.

Now you have completed a discrete CMOS inverter design, with its own symbol, and the instance parameters as specified. The next step is to perform a simulation on the inverter to verify that it works and to evaluate its performance.

8 Creating Parameterized Inverter

In the previous example, we created inverter schematic and symbol with fixed sizes (wp/lp = 390n/45n, wn/ln = 260n/45n). What if you need another inverter with different sizes? You can create new inverter with different sizes (this is what is done in a standard digital cell library), but it is often easier to use a generalized inverter cell that you can specify the sizes in the higher level design while doing your initial design. In this section, you will generate an inverter cell and symbol with the sizes parameterized.

8.1 *Creating Schematic*

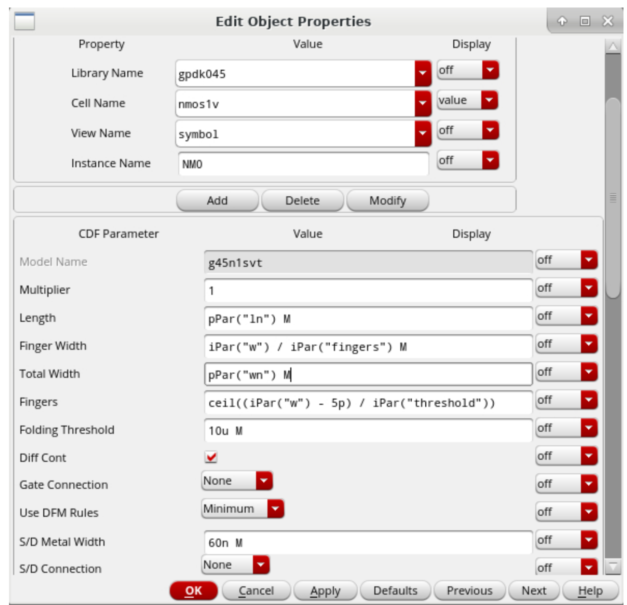
Creating a parameterized cell view is basically the same as the previous example except property setting.

- Copy your current inverter by going into the library manager, selecting your inverter, then doing right click -> copy. Change the “To” cell to be “inverterp”

- Open the inverterp schematic

- In the property window of pmos1v (click pmos1v and use short cut key ‘q’), set width and length as follows

    - Total width : **pPar(“wp”)**
    - length: **pPar(“lp”)**



- Note that many equations will auto-populate in the other fields. These equations will auto-calculate the layout parameters based on your total width and length.

- In the property window of nmos1v (click nmos1v and use short cut key ‘q’), set width and length as follows

    - Total width : **pPar(“wn”)**
    - length: **pPar(“ln”)**

- Check and Save the schematic

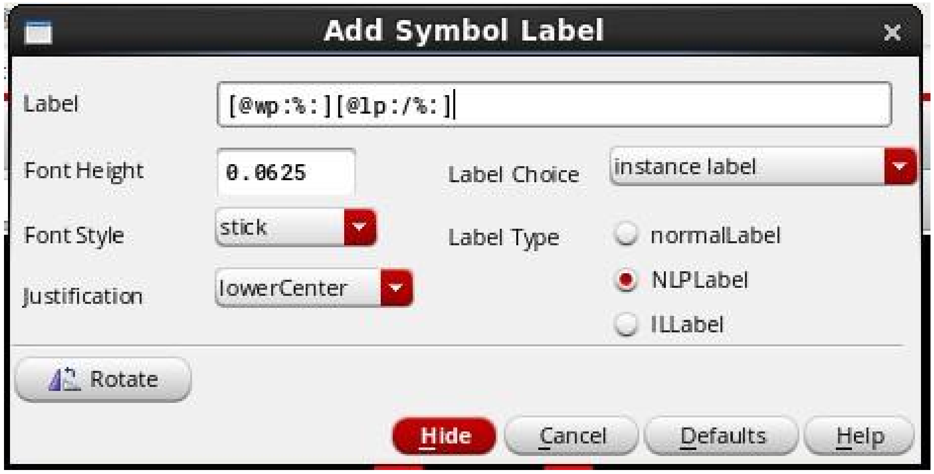
8.2 *Creating Symbol*

- Return to the library manager and open the symbol for your inverterp cell.

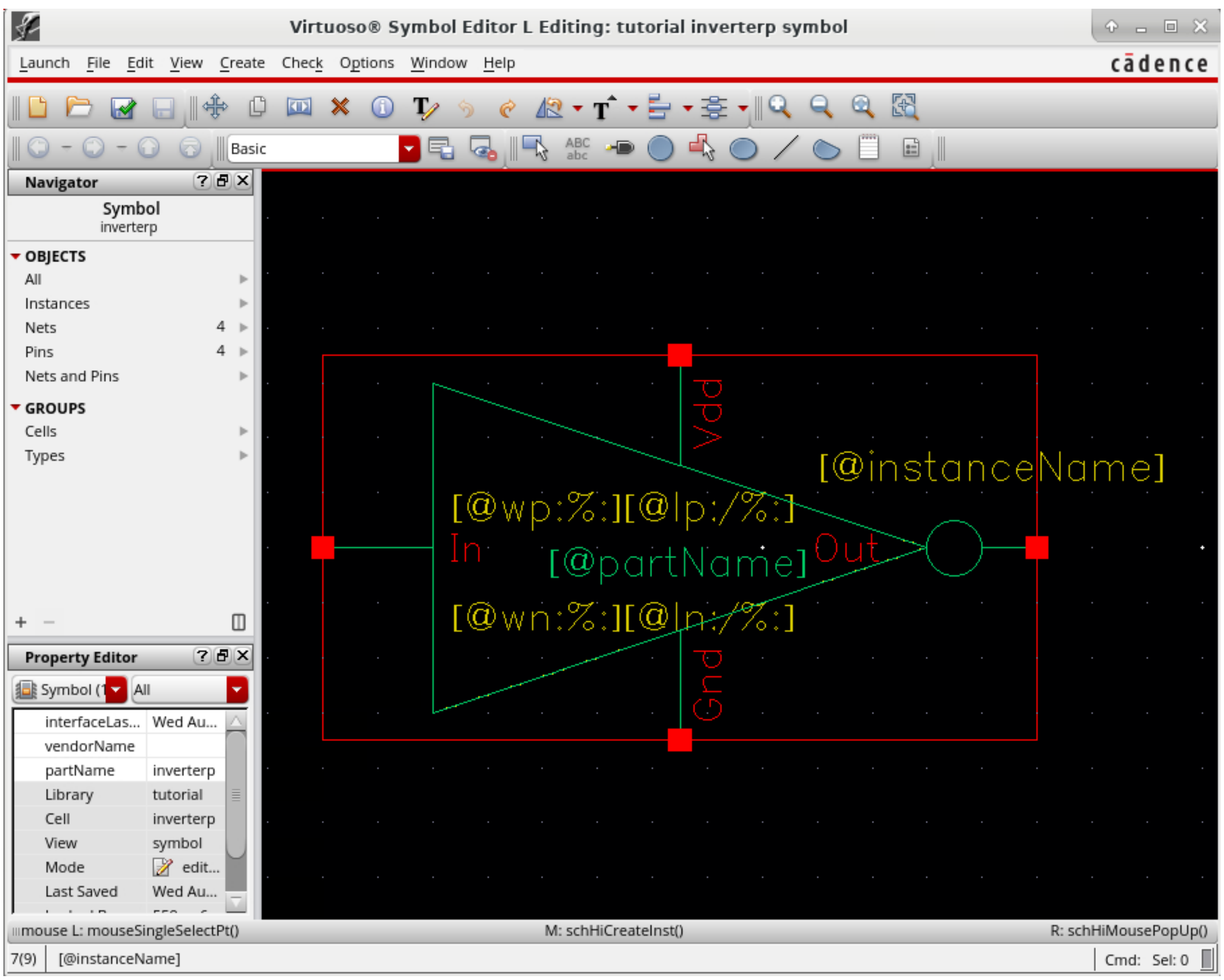
- To add symbol label, select **Create -› Label** and set the options as shown in the following picture.

- Repeat adding symbol label for *nmos4* with the Label field ‘**[@wn:%:][@ln:/%:]**’.

- Check and Save the symbol.

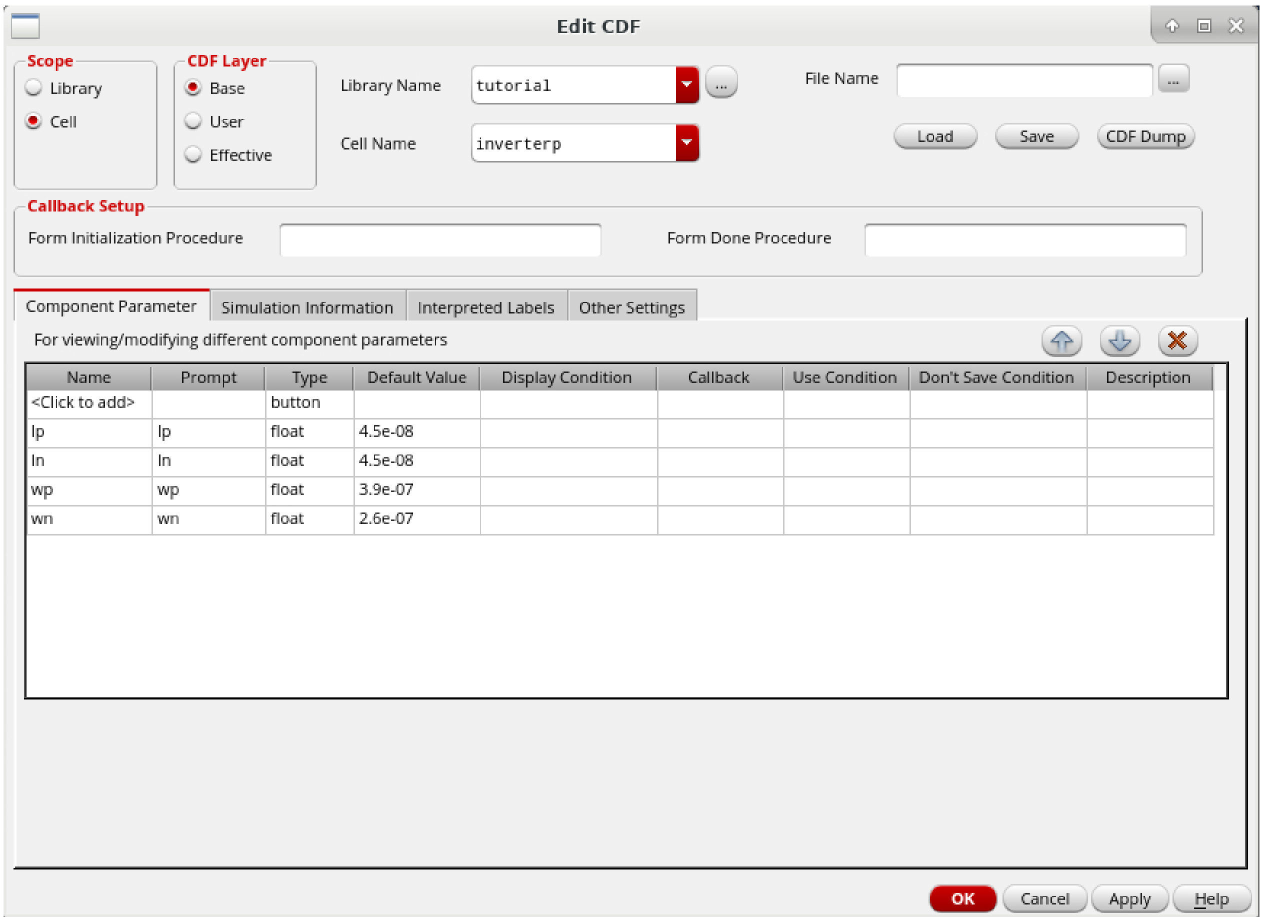


- The final picture should look like the figure below.



8.3 Adding CDF parameter

- In the CIW, select **Tools -› CDF -› Edit…**
- Set Scope to *Cell*, CDF Layer to *Base*.
- *Library Name* should be set to *tutorial*, *Cell Name* to *inverterp* (use drop-down menus to select the library and cell name).

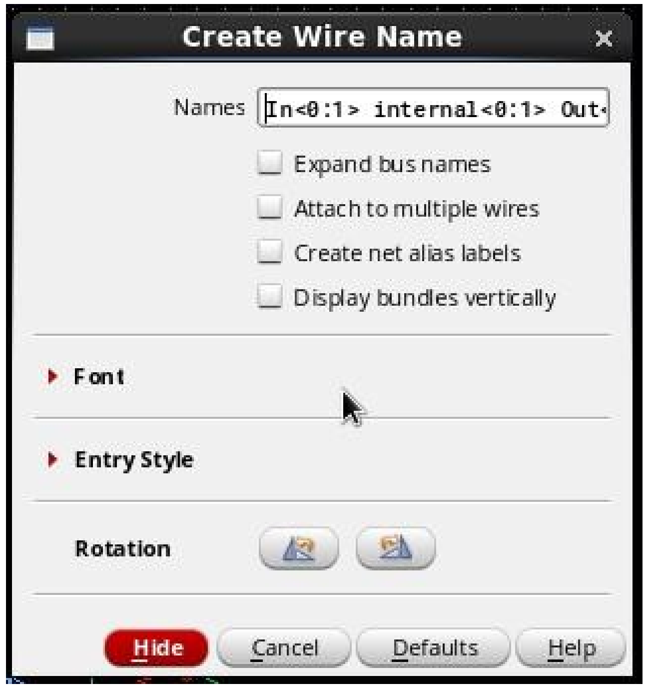


- For *wp* parameter, set *Type* to **float** (double click on the field to edit), *Store Default* to **yes**, *Name* to **wp**, *Prompt* to **wp** and *Default Value* to the default value you want (for example, for **390n**, type **390e-9**).
- Repeat adding CDF parameters for *lp, wn* and *ln*.
- Now the parameterized inverter is ready to be used.

9 Create a New Schematic to be simulated

9.1 *InverterTest Schematic*

- Under the same library (tutorial), create a new cellview named *InverterTest*.
- Use the *Add Instance* command to place your *inverterp* symbol in the cellview. It should be found under the *tutorial* library.
- If you open the property window of *inverterp*, you can find 4 CDF parameters. If the values for those parameters are not specified, set those parameters to *wp/lp =540n/180n* and *wn/ln = 270n/180n*.
- Use the *Add Instance* command to place two capacitors in the cellview from the *NCSU_Analog_Parts* library and give them a value of `10fF (when you type in the value in the property window, just type 10f).`
- For this tutorial, we will create two bit wide buses for input, internal nodes and output. To add a bus, click the *Wire (wide)* icon. Draw the bus as you would draw any normal (narrow) wire.
- After creating the bus, the bus must be named. The same command to add wire names for narrow wires is used, however the syntax for naming the bus is slightly different.
- Bus names is in the form *`Name<a:b>`*, where `a` and `b` denotes the range of bits of the bus. Several two bit buses that will be created in this tutorial include `In<0:1>`, `internal<0:1>` and `Out<0:1>`.
- Similar to the previous tutorial, it is possible to add several names to different buses at one time. To do this, make sure the *Expand bus names* button is off and simply type the names of the buses separated by a space in the *Names* field. The names will be added in order when you click on the various nets. As an exercise, create and name the buses similar to the ones in the final schematic.



- To draw or tap wires to and from the buses, draw wires from the buses and name them correspondingly to their bits. To name the individual bit lines, type in the bus name (e.g. `In<0:1>` or simply `<0:1>`) and turn on *Bus Expansion* in the *Add Wire Name* window. The *Bus Expansion* button is used to extract individual bit names from the bus. When you start placing the names on the wires, you will notice that the first name will be `In<0>` (or `<0>`), the next will be `In<1>` (or `<1>`), and so on.

- Setting the *Placement* button to multiple allows you to place an array of names, rather than one at a time.

- Complete, check and save the schematic so that it looks like the schematic shown below. Don't forget to add the pins `In<0:1>` and `Out<0:1>`. Note that vdd instances and gnd instances are added (from *Supply_Nets* under *analogLib* library) with connections as shown in the figure.

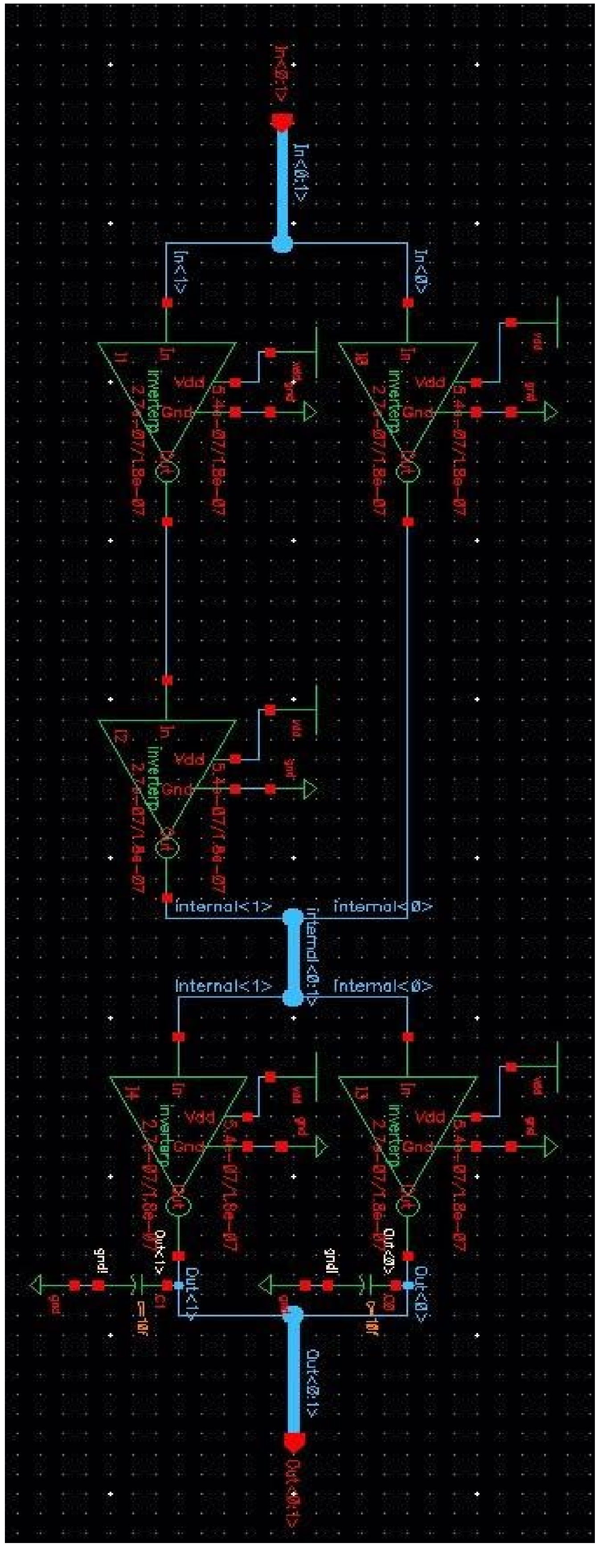
9.2 Hierarchy Editing

- Suppose that you wanted to explore the schematic of the inverter. You can traverse a design hierarchy to view or edit your inverter design by selecting the inverter, then select **Edit -› Hierarchy -› Descend Edit / Read (hotkeys are E/ e)**. Select *schematic* for the *View Name*. Click **OK** in the new window that appears.

- Now the cellview should change to the schematic view of your inverter, showing the transistors and the connections. If there are more levels to descend, you can continue to do so from here.

- Note that if you make and save any changes to the schematic in the lower levels, the whole design (top level) will be changed, including other designs that use this symbol.

- To return up one level or to the top view, use **Edit -› Hierarchy -› Return /Return To Top (hotkey is B)**.


In<0:1>
In<1>
In<0>
I0
I1
I2
I3
I4
inverterp
In
Vdd
Gnd
Out
vdd
gnd
2.7e-07/1.8e-07
5.4e-07/1.8e-07
internal<1>
internal<0>
internal<0:1>
Out<1>
Out<0>
C1
C0
c=10f
Out<0:1>